

BUSAN GIMHAE INTL, Korea, Republic of

WMO: 471530

Lat: 35.180N

Lon: 128.938E

Elev: 2

StdP: 101.30

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.1	(c) -4.9	(d) -19.1	(e) 0.7	(f) -1.0	(g) -17.0	(h) 0.8	(i) -0.7	(j) 9.7	(k) 2.3	(l) 8.6	(m) 3.3	(n) 3.1	(o) 330	(p) 0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.9	(c) 33.0	(d) 25.9	(e) 31.9	(f) 25.5	(g) 30.2	(h) 24.8	(i) 26.8	(j) 30.8	(k) 26.3	(l) 30.2	(m) 25.7	(n) 29.4	(o) 4.7	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	21.3	29.6	25.1	20.2	28.8	24.2	19.2	28.1	84.7	31.2	81.9	30.3	78.9	29.4	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.0	7.8	7.0	-9.1	34.8	1.7	1.6	-10.3	36.0	-11.3	36.9	-12.3	37.8	-13.5	39.0		
			-10.7	27.8	1.5	1.1	-11.8	28.6	-12.7	29.3	-13.6	29.9	-14.7	30.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	2.4	4.5	8.5	13.7	18.6	22.2	25.9	27.1	22.8	17.3	10.6	4.4		
	DBStd	8.83	3.10	3.34	3.12	2.83	2.30	1.76	2.41	2.18	2.42	2.90	3.63	3.40		
	HDD10.0	678	237	157	66	4	0	0	0	0	0	1	37	177		
	HDD18.3	2071	495	388	304	139	26	1	0	0	1	53	232	433		
	CDD10.0	2456	1	2	20	117	265	366	492	529	383	226	55	2		
	CDD18.3	808	0	0	0	2	32	116	233	271	133	21	0	0		
	CDH23.3	6206	0	0	0	11	167	503	1961	2702	777	84	1	0		
	CDH26.7	1879	0	0	0	1	18	68	622	1004	164	3	0	0		
(14) Wind	WSAvg	3.3	3.2	3.3	3.5	3.6	3.4	3.3	3.4	3.3	3.3	2.9	2.8	3.0		
(15) Precipitation	PrecAvg	1447	27	45	75	125	143	179	292	268	156	60	49	27		
	PrecMax	2379	63	113	149	233	310	342	680	723	476	188	210	110		
	PrecMin	824	0	0	7	27	45	45	82	57	23	1	0	0		
	PrecStd	349	17	29	38	57	62	96	154	176	122	47	44	28		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	16.2	20.1	25.0	28.2	30.0	34.8	35.0	31.2	26.9	22.2	16.2		
		MCWB	10.0	11.2	11.6	16.0	18.2	21.0	26.5	26.1	24.4	19.1	15.8	11.3		
	2%	DB	11.2	14.2	17.9	22.8	26.9	28.2	32.9	33.2	29.9	25.2	20.1	14.1		
		MCWB	6.1	9.4	11.5	14.7	17.6	20.6	26.0	25.9	23.3	18.4	14.6	9.0		
	5%	DB	9.8	12.8	16.1	21.0	25.1	27.2	31.2	32.2	28.8	24.1	18.8	12.2		
		MCWB	5.1	8.0	10.7	14.2	17.2	20.4	25.6	25.6	22.6	18.1	13.6	7.6		
	10%	DB	8.0	10.9	14.9	19.8	23.9	26.1	30.0	31.1	27.2	22.9	17.0	10.8		
		MCWB	4.1	6.1	10.1	13.8	16.9	20.3	25.2	25.4	21.6	17.0	12.5	6.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	14.0	15.2	19.0	21.4	24.3	27.7	28.0	26.1	22.5	18.0	12.9		
		MCDB	13.2	15.9	17.2	21.1	24.3	26.9	31.8	31.6	29.7	24.6	20.1	14.9		
	2%	WB	7.5	10.9	13.7	17.2	19.9	23.3	26.7	27.1	24.9	20.6	16.1	10.3		
		MCDB	9.8	13.2	16.1	20.2	23.1	25.6	30.7	31.1	28.0	23.6	19.0	13.1		
	5%	WB	5.9	8.2	12.0	15.8	19.0	22.5	26.2	26.5	24.0	19.2	14.7	8.4		
		MCDB	8.5	11.5	15.3	19.0	22.7	25.0	30.0	30.5	26.8	22.8	17.8	11.4		
	10%	WB	4.4	6.6	10.3	14.7	18.2	21.7	25.6	26.0	22.7	17.7	13.1	6.8		
		MCDB	7.4	9.8	14.0	18.1	22.0	24.5	29.2	29.9	25.9	21.6	16.2	10.0		
(35) Mean Daily Temperature Range	MDBR		9.9	10.3	10.3	10.3	9.5	7.6	6.2	6.9	8.1	10.1	10.6	10.3		
	5% DB	MCDBR	11.8	12.3	12.1	12.5	12.0	9.7	8.0	8.4	9.2	11.0	12.0	12.3		
		MCWBR	7.9	8.3	7.6	7.0	5.9	4.7	3.8	3.6	4.4	5.7	7.3	8.4		
	5% WB	MCDBR	9.7	10.9	10.2	9.5	9.3	6.3	7.1	7.4	7.2	8.8	10.2	11.0		
		MCWBR	7.5	8.3	7.9	6.7	5.8	4.0	3.8	3.7	4.3	5.6	7.4	8.3		
(40) Clear-Sky Solar Irradiance	taub		0.410	0.461	0.536	0.567	0.580	0.620	0.592	0.548	0.496	0.461	0.429	0.403		
	taud		2.082	1.981	1.820	1.800	1.822	1.802	1.931	2.045	2.135	2.113	2.110	2.114		
	Ebn at Noon		756	759	735	739	736	705	722	746	766	753	730	735		
	Edh at Noon		126	157	200	214	212	216	189	166	144	135	122	116		
(44) All-Sky Solar Radiation	RadAvg		2.69	3.32	4.32	5.00	5.49	4.85	4.59	4.66	3.90	3.61	2.75	2.45		
	RadStd		0.21	0.36	0.40	0.39	0.58	0.44	0.70	0.57	0.41	0.34	0.33	0.15		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+60	N/A

Nomenclature: See separate page